

ChIP-seq Data Analysis Report

Overview

Chromatin immunoprecipitation sequencing (ChIP-seq) is a method that allows researchers to map where various transcription regulators are located at a genome-wide level. It has and continues to be a crucial tool in molecular biology and genomics, providing valuable insights into the dynamic nature of the epigenetic landscape in various biological contexts and diseases. Active Motif's ChIP-Seq service provides a comprehensive solution for profiling transcription factor and histone modifications genome-wide.

QC

Differential Peak Analysis

UCSC Genome Browser

Pipeline Information

Data Explanation

[Download and Access Data](#)

Deliverables

Active Motif's standard analysis provides a number of popular processed data files, as well as numerous results and graphics. The following folders are provided to the customer:

1.

00_Raw_Fastqs: This folder contains raw fastq files straight off the sequencer. No processing has been done on these fastqs. (Always provided)
2.

01_QC: This folder contains a number of qc related files and plots that can be used to determine how well your ChIP-seq experiment worked. (Always provided)
3.

02_Alignment: This folder contains the aligned and filtered BAM files along with their generated index files. (Always provided)
4.

03_Bigwigs: This folder contains binary, compressed and indexed bigWig files that have been normalized using either Counts Per Million (CPM) normalization or spike-in normalized. (Always provided)
5.

04_HOMER: This folder contains HOMER tag directories which include useful information regarding the count and length distribution of your reads. Tag directories are also the most commonly required input for any HOMER suite analysis. (Always provided)
6.

05_UCSC_Browser_TrackHubs: This folder contains text files with UCSC Genome Browser trackhub addresses for easily uploading and visualizing your ChIP-seq data. See the [UCSC Genome Browser](#) tab for more information. (Always provided)
7.

06_Peak_Analysis: This folder contains information about the peak calls for your samples including annotated and non-annotated peak files, frip scores, and consensus peak files. (Always provided)
8.

07_Differential_Peak_Analysis: This folder contains information about any differential peak analysis carried out including raw counts, edgeR results, motif analysis, and a number of plots that can be used for visualization. (Only provided when differential peak analysis is carried out)

00_Raw_Fastqs/ Folder

Files

1.

Raw FASTQ files with extension *.fastq.gz . Paired-end reads typically contain `_(R1|r1)` and `_(R2|r2)` in the filename while single-end reads do not.

01_QC/ Folder

Subdirectories

1.

fastqc/ - Contains FastQC .html files for raw reads.
2.

homer_motifs/ - Contains folders that contain motif analysis results for every sample. `knownResults.txt` and `knownResults.html` show the enrichment statistics of known transcription factor motifs while `homerResults.txt` and `homerResults.html` show the enrichment statistics of de novo motifs found as well as what known transcription factor motif it most closely resembles.
3.

multiqc_data/ - Contains all the text files used to generate multiqc tables and plots in the `multiqc_report.html`.
4.

phantompeakqualtools/ - Contains informative enrichment and quality measures such as NSC and RSC scores as computed by the phantompeakqualtools software. Also contains plots of the cross-correlation values.
5.

plots/ - Contains a number of plots including sample-to-sample correlation plots, fingerprint plots, heatmaps at transcription start sites, motif enrichment dotplots, and genome-wide bin PCA plots.
6.

samtools_stats/ - Contains read information generated by various samtools commands for all samples after read filtering has been performed.

Files

1.

multiqc_report.html - MultiQC html report that contains several major quality control metrics and plots in a single easy to visualize format.
2.

qc_metrics.mqc.stats - Csv file that contains sequencing and qc statistics. Same as in table at top of `multiqc_report.html` file.

02_Alignment/ Folder

Files

1.

Filtered, deduplicated and sorted BAM files (`.bam`) and their associated indexes (`.bai`). BAM files are a binary version of SAM files that include information about where reads are aligning on the genome.

03_Bigwigs/ Folder

Files

1.

Counts per Million (CPM) and/or Spike-in normalized bigWig files. bigWig files are binary versions of bedGraph files that include information on the number of reads that overlap a genomic region bin. By default our bin-sizes are 10-bp.

04_HOMER/ Folder

Subdirectories

1.

tag_directories/ - Contains HOMER generated tag directories for each sample. Tag directories are a HOMER-specific data structure that includes information about where reads are mapping on the genome as well as basic qc stats. These are extremely useful for carrying out your own analysis using the HOMER suite.

05_UCSC_Genome_Brower_TrackHubs/

Files

1.

trackhubs_[antibody].txt - Text file(s) that contain(s) the address information required to create a TrackHub on the UCSC Genome Browser of your samples. Our analysis generates a single text file per antibody.

06_Peak_Analysis/ Folder

Subdirectories

1.

annotated_peaks/ - Contains HOMER-annotated peak files. Annotated peaks include information about a potential binding site's nearest gene and the genomic location of that peak.
2.

consensus_peak_analysis/ - Contains consensus peak files and plots where all antibody peaks are overlapping (`all_antibodies/`) or antibody-specific peaks are overlapping (`per_antibody/`).
3.

frip_scores/ - Contains text files that include the fraction of reads in peaks per sample.
4.

macs2/ - Contains MACS2 `.narrowPeak` and/or `.broadPeak` files. (Only provided for narrow or broad peak shapes)
5.

epic2/ - Contains EPIC2 `.broadPeak` files. (Only provided for very-broad peak shapes)

07_Differential_Peak_Analysis/ Folder

Subdirectories

1.

edger/ - Contains differential analysis information generated using edgeR and is made up of multiple subdirectories: `differential_analysis` and `graphics`. The `differential_analysis` folder includes the differential analysis results for each antibody and for each condition tested (please see the the METADATA table provided in the **Differential Peak Analysis** tab for information about what conditions were tested in your analysis). The `graphics` folder gives includes several quality control plots that include all samples.
2.

meirlop_motifs/ - Contains the differential motif analysis carried out by MEIRLOP. The results of the analysis can be found in both text `lr_results.tsv` and html `lr_results.html` format. Information about what peaks contain what motifs can be found in `scan_results.tsv`.
3.

raw_counts/ - Contains raw peak count text files generated either by FeatureCounts.
4.

scored_fasta_files/ - Contains the Log2 Fold Change scored fasta files that are used as input by MEIRLOP for differential motif analysis.

If you have any questions about the laboratory processing of the samples in this project, please contact our Epi-Services team at services_lab@activemotif.com. If you have any questions about the enclosed analysis, please reach out to our Bioinformatics team at services_informatics@activemotif.com.